

SORTING

PROBLEM 7

QUESTION:

DS	Session	Sorting	Question Information	Level 1	Challenge 17
Question Description: Simon is studying B.Tech.-Mechanical Engineering. He's going to attend a computer science-based subject exam this semester. Due to the less preparation in the previous monthly tests, his internal mark decreased. His computer science Professor made an offer one more chance to boost up his internal marks. Professor assigns a program to Simon for the internal mark bootup. So Simon wants to solve the given task is, Given two arrays, A and B, of equal size n, the task is to find the minimum value of $A[0] * B[0] + A[1] * B[1] + \dots + A[n-1] * B[n-1]$, where shuffling of elements of arrays A and B is allowed. can you help him in solving Questions ? Constraints: $1 \leq T \leq 100$ $1 \leq N \leq 50$ $1 \leq A[i] \leq 20$ Input Format:					

The first line of input contains an integer denoting the no of test cases.

Then T test cases follow. Each test case contains three lines.

The first line of input contains an integer N denoting the size of the arrays.

In the second line are N space separated values of the array A[], and in the last line are N space separated values of the array B[].

Output Format:

For each test case in a new line print the required result.

Example :

Input : A[] = {3, 1, 1} and B[] = {6, 5, 4}.

Output : 23 Minimum value of $S = 1 * 6 + 1 * 5 + 3 * 4 = 23$.

Input : A[] = { 6, 1, 9, 5, 4 } and B[] = { 3, 4, 8, 2, 4 }

Output : 80. Minimum value of $S = 1 * 8 + 4 * 4 + 5 * 4 + 6 * 3 + 9 * 2 = 80$.

Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN)	INPUT (STDIN)
3	2
4	3
2 5 3 7	13 11 1
8 5 7 3	6 15 14
3	5
13 11 1	6 11 9 15 4
6 15 14	3 41 8 21 4
5	
6 11 9 15 4	EXPECTED OUTPUT
3 41 8 21 4	
	247
	451

EXPECTED OUTPUT 83 247 451		
▼ Mandatory Test Cases Test Case 1 KEYWORD void sort(int a[],int n,int flag)		
▼ Complexity Test Cases Test Case 1 CYCOMATIC COMPLEXITY 1 Test Case 2 TOKEN COUNT 428 Test Case 3 NLOC 55		

SOURCE CODE:

```
#include <stdio.h>
#include <stdlib.h>

int cmp_up(const void* a, const void* b) {
    return (*(const int*)a - *(const int*)b);
}

int cmp_down(const void* a, const void* b) {
    return (*(const int*)b - *(const int*)a);
}

int main() {
    int t_cases;
    if (scanf("%d", &t_cases) != 1) return 0;

    while (t_cases--) {
        int length;
        scanf("%d", &length);

        int arr_x[length], arr_y[length];
        for (int i = 0; i < length; i++) scanf("%d", &arr_x[i]);
        for (int i = 0; i < length; i++) scanf("%d", &arr_y[i]);

        qsort(arr_x, length, sizeof(int), cmp_up);
        qsort(arr_y, length, sizeof(int), cmp_down);

        long long product_sum = 0;
        for (int i = 0; i < length; i++) {
            product_sum += (long long)arr_x[i] * arr_y[i];
        }
        printf("%lld\n", product_sum);
    }
    return 0;
}
```

OUTPUT:

```
2
3
13 11 1
6 15 14
5
6 11 9 15 4
3 41 8 21 4
247
451
```

```
3
4
2 5 3 7
8 5 7 3
3
13 11 1
6 15 14
5
6 11 9 15 4
3 41 8 21 4
83
247
451
```